

Operations Research

Problem Set G1: Graph Models (Foundations)

Companion to Chapter 1, Section 1.6

Scope. Incidence sets and degrees, paths and Euler trails, and the four graph models of Chapter 1: shortest path, maximum flow, minimum-cost flow and assignment. In every problem, “prove optimal” means: give a feasible plan *and* a bound that no plan can beat, as in Section 1.5. Solutions start on page 4.

Problem 1 (Incidence sets, degrees, Euler trails)

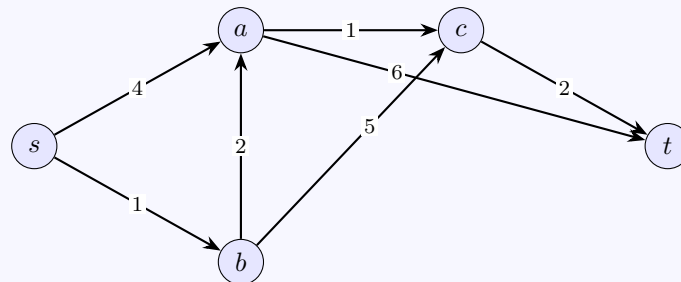
Six sensor sites are linked by seven candidate cables. The undirected graph $G = (V, E)$ has $V = \{1, \dots, 6\}$ and

$$E = \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{4, 6\}, \{5, 6\}\}.$$

- Draw G . Write $\delta(v)$ and $\deg v$ for every vertex, and verify the handshake lemma numerically.
- For $S = \{1, 2, 3\}$ and for $S = \{3, 4\}$, list $\delta(S)$ and $E(S)$. Which of the two groups is “more isolated,” and in what sense?
- A technician must inspect every cable exactly once, walking along the cables. Is this possible? If so, give the walk and say where it must start and end; if not, explain why.
- The technician would prefer to start and finish at the same site. Show that this is impossible in G . Then show that *removing* cable $\{3, 4\}$ makes every degree even and yet still does not allow such a closed walk. Which hypothesis of Euler’s theorem fails?

Problem 2 (Shortest path: fastest delivery)

A courier in Casablanca must carry a parcel from a depot s to a client t . The one-way street network is the digraph below; each arc carries a travel time in minutes.



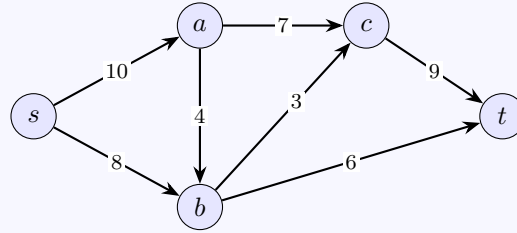
- Write N , A , $\delta^+(b)$ and $\delta^-(a)$. Explain why the route $s \rightarrow a \rightarrow b \rightarrow c \rightarrow t$ is not allowed.
- List every directed s – t path with its total time. Which is fastest?
- Write the shortest-path integer program (variables $x_{ij} \in \{0, 1\}$, one balance equation per node, objective in minutes). Give the vector x of the fastest route and check every constraint.
- Prove* that no route is faster, without listing paths: find numbers d_s, d_a, d_b, d_c, d_t with

$d_s = 0$ and $d_j \leq d_i + w_{ij}$ for every arc (i, j) , and show that any s - t path has length at least d_t . (Hint: add the inequalities along the path.)

Problem 3 (Maximum flow: a water network)

A water utility pumps from a reservoir s to a town t through pumping stations a, b, c . Pipe capacities are in thousands of m^3 per hour:

$$u_{sa} = 10, \quad u_{sb} = 8, \quad u_{ab} = 4, \quad u_{ac} = 7, \quad u_{bc} = 3, \quad u_{bt} = 6, \quad u_{ct} = 9.$$

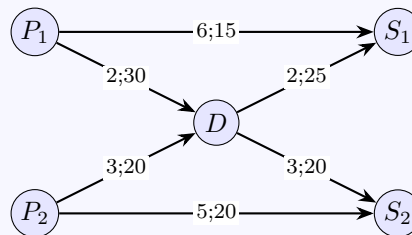


- Draw the network and write the maximum-flow linear program: variables, objective, capacity constraints, and one conservation equation for each of a, b, c .
- The operator proposes $x_{sa} = 7, x_{sb} = 8, x_{ab} = 0, x_{ac} = 7, x_{bc} = 2, x_{bt} = 6, x_{ct} = 9$. Check that this is a feasible flow and compute its value. Which pipes are saturated?
- Find an s - t cut S whose capacity $u(\delta^+(S))$ equals the value in (b), and conclude that the proposed flow is maximal. State precisely which theorem you are using.
- The utility can afford one upgrade. Option 1: enlarge (s, a) from 10 to 20. Option 2: enlarge (c, t) from 9 to 12. For each option, what is the new maximum flow? Justify each answer with a flow and a cut.

Problem 4 (Minimum-cost flow: fertiliser through a depot)

Two plants P_1 and P_2 hold 30 and 20 tonnes of fertiliser. Two shops S_1 and S_2 have ordered 25 tonnes each. Deliveries may go directly or through a depot D . Each arc is labelled (cost in hundreds of MAD per tonne; capacity in tonnes):

Arc	Cost	Capacity
(P_1, D)	2	30
(P_2, D)	3	20
(P_1, S_1)	6	15
(D, S_1)	2	25
(D, S_2)	3	20
(P_2, S_2)	5	20



- Draw the network, give the balance b_i of every node, and check that the balances sum to zero. Why must they?
- Write the minimum-cost flow linear program in full (six variables, five balance equations, six capacity bounds).
- For each shop, list every route from a plant and its cost per tonne. Propose a plan, check it against every constraint, and compute its cost.
- Prove your plan is optimal by a relaxation argument: drop the supply and capacity constraints and bound the cost per tonne delivered to each shop.

- (e) Plant P_1 reports that only 25 tonnes are actually in stock. Show, without solving anything, that no feasible plan exists.

Problem 5 (Assignment: analysts and incidents)

Three analysts A, B, C must each handle exactly one of three incidents 1, 2, 3. Estimated handling times (hours):

	1	2	3
A	1	2	9
B	2	4	9
C	9	9	3

- Write the assignment model with nine binary variables: objective, three “each analyst” constraints, three “each incident” constraints.
- A manager assigns greedily: repeatedly pick the cheapest remaining (analyst, incident) pair. What assignment results, and what is its total time?
- Enumerate all six complete assignments and find the optimum. Is the greedy answer optimal?
- Explain why “sum of the row minima” is a lower bound on the optimum (which constraints does it relax?). Compute it. Does it prove optimality of your answer in (c)?
- Show that if numbers u_A, u_B, u_C and v_1, v_2, v_3 satisfy $u_i + v_j \leq c_{ij}$ for every pair, then every assignment costs at least $\sum_i u_i + \sum_j v_j$. Find such numbers proving that the optimum in (c) cannot be improved.

Problem 6 (Short proofs)

- In a digraph, show that $\sum_{i \in N} |\delta^+(i)| = \sum_{i \in N} |\delta^-(i)| = |A|$.
- In the maximum-flow model, suppose someone adds the conservation equation $x(\delta^+(s)) = x(\delta^-(s))$ at the source as well. Show that the only feasible flows then have value 0.
- Let x be a flow and S an s - t cut such that every arc leaving S is saturated ($x_{ij} = u_{ij}$) and every arc entering S carries zero flow. Prove that x is a maximum flow.
- In the minimum-cost flow model, suppose the plant constraints are inequalities $x(\delta^+(i)) - x(\delta^-(i)) \leq b_i$ for supply nodes and the demand constraints remain equalities. Show that feasibility requires total supply $\geq$ total demand.

Solutions

Solution 1

- (a) $\delta(1) = \{12, 13\}$, $\delta(2) = \{12, 23\}$, $\delta(3) = \{13, 23, 34\}$, $\delta(4) = \{34, 45, 46\}$, $\delta(5) = \{45, 56\}$, $\delta(6) = \{46, 56\}$ (writing ij for $\{i, j\}$). Degrees 2, 2, 3, 3, 2, 2; sum = 14 = $2 \cdot 7 = 2|E|$.
- (b) $S = \{1, 2, 3\}$: $\delta(S) = \{34\}$, $E(S) = \{12, 13, 23\}$. $S = \{3, 4\}$: $\delta(S) = \{13, 23, 45, 46\}$, $E(S) = \{34\}$. The group $\{1, 2, 3\}$ is more isolated: a single cable connects it to the rest, so cutting $\{3, 4\}$ disconnects it.
- (c) Exactly two vertices (3 and 4) have odd degree, so by Euler's theorem a walk using every edge once exists, and it must start at one of them and end at the other. Example: 3, 1, 2, 3, 4, 5, 6, 4 uses 13, 12, 23, 34, 45, 56, 46.
- (d) A closed walk needs all degrees even; vertices 3, 4 are odd, so none exists. Removing $\{3, 4\}$ gives degrees 2, 2, 2, 2, 2, 2, but the graph splits into two components $\{1, 2, 3\}$ and $\{4, 5, 6\}$: the *connectedness* hypothesis fails, and no single walk can cover both triangles.

Solution 2

- (a) $N = \{s, a, b, c, t\}$, $A = \{(s, a), (s, b), (b, a), (a, c), (b, c), (a, t), (c, t)\}$. $\delta^+(b) = \{(b, a), (b, c)\}$, $\delta^-(a) = \{(s, a), (b, a)\}$. The route $s \rightarrow a \rightarrow b$ would use (a, b) , which is not an arc; only (b, a) exists, and arcs must be traversed in their direction.
- (b) Five paths: sat : 10; $sact$: 7; $sbat$: 9; $sbact$: 6; $sbcct$: 8. Fastest: $s \rightarrow b \rightarrow a \rightarrow c \rightarrow t$ in 6 minutes.
- (c) Minimise $4x_{sa} + x_{sb} + 2x_{ba} + x_{ac} + 5x_{bc} + 6x_{at} + 2x_{ct}$ subject to

$$\begin{aligned} x_{sa} + x_{sb} &= 1 & (s) \\ x_{ac} + x_{at} - x_{sa} - x_{ba} &= 0 & (a) \\ x_{ba} + x_{bc} - x_{sb} &= 0 & (b) \\ x_{ct} - x_{ac} - x_{bc} &= 0 & (c) \\ -x_{at} - x_{ct} &= -1 & (t) \end{aligned}$$

and $x_{ij} \in \{0, 1\}$. The fastest route is $x_{sb} = x_{ba} = x_{ac} = x_{ct} = 1$, others 0: at s : $0 + 1 = 1$; at a : $1 + 0 - 0 - 1 = 0$; at b : $1 + 0 - 1 = 0$; at c : $1 - 1 - 0 = 0$; at t : $-0 - 1 = -1$. Objective $1 + 2 + 1 + 2 = 6$.

- (d) Take $d_s = 0$, $d_b = 1$, $d_a = 3$, $d_c = 4$, $d_t = 6$. Check each arc: (s, a) : $3 \leq 0 + 4$; (s, b) : $1 \leq 0 + 1$; (b, a) : $3 \leq 1 + 2$; (a, c) : $4 \leq 3 + 1$; (b, c) : $4 \leq 1 + 5$; (a, t) : $6 \leq 3 + 6$; (c, t) : $6 \leq 4 + 2$. For any path $s = v_0, v_1, \dots, v_k = t$, adding $d_{v_i} - d_{v_{i-1}} \leq w_{v_{i-1}v_i}$ over $i = 1, \dots, k$ telescopes to $d_t - d_s \leq \text{length}$, i.e. $\text{length} \geq 6$. The path in (b) has length 6, so it is optimal.

Solution 3

- (a) Maximise $x_{sa} + x_{sb}$ subject to $0 \leq x_{ij} \leq u_{ij}$ on all seven arcs and

$$x_{ab} + x_{ac} = x_{sa} \quad (a), \quad x_{bc} + x_{bt} = x_{sb} + x_{ab} \quad (b), \quad x_{ct} = x_{ac} + x_{bc} \quad (c).$$

- (b) Capacities: $7 \leq 10$, $8 \leq 8$, $0 \leq 4$, $7 \leq 7$, $2 \leq 3$, $6 \leq 6$, $9 \leq 9$. Conservation: a : $0 + 7 = 7$; b : $2 + 6 = 8 + 0$; c : $9 = 7 + 2$. Value $7 + 8 = 15$. Saturated: (s, b) , (a, c) , (b, t) , (c, t) .

- (c) $S = \{s, a, b, c\}$: $\delta^+(S) = \{(b, t), (c, t)\}$, capacity $6 + 9 = 15$. By the cut bound (Theorem 1.6.12: every flow value is at most every cut capacity), no flow exceeds 15; the flow in (b) reaches 15, so it is maximal (matching bounds).
- (d) *Option 1*: the cut $\{s, a, b, c\}$ still has capacity 15, so the maximum stays 15: the upgrade is useless. *Option 2*: the cut $\{s, a, b, c\}$ becomes $6 + 12 = 18$, but the cut $\{s, a, b\}$ has $\delta^+ = \{(a, c), (b, c), (b, t)\}$ with capacity $7 + 3 + 6 = 16$, so the maximum is at most 16. A flow of 16: $x_{sa} = 8$, $x_{sb} = 8$, $x_{ab} = 1$, $x_{ac} = 7$, $x_{bc} = 3$, $x_{bt} = 6$, $x_{ct} = 10$ (check b : $3 + 6 = 8 + 1$; c : $10 = 7 + 3$; $x_{ct} = 10 \leq 12$). New maximum 16.

Solution 4

- (a) $b_{P_1} = 30$, $b_{P_2} = 20$, $b_D = 0$, $b_{S_1} = -25$, $b_{S_2} = -25$; sum = 0. Summing all balance equations, each arc variable appears once with + and once with -, so the left side is 0; hence $\sum_i b_i$ must be 0 for any feasible flow to exist.
- (b) With $x_1 = x_{P_1D}$, $x_2 = x_{P_2D}$, $x_3 = x_{P_1S_1}$, $x_4 = x_{DS_1}$, $x_5 = x_{DS_2}$, $x_6 = x_{P_2S_2}$:

$$\min 2x_1 + 3x_2 + 6x_3 + 2x_4 + 3x_5 + 5x_6$$

s.t. $x_1 + x_3 = 30$, $x_2 + x_6 = 20$, $x_4 + x_5 - x_1 - x_2 = 0$, $-x_3 - x_4 = -25$, $-x_5 - x_6 = -25$, and $0 \leq x_1 \leq 30$, $0 \leq x_2 \leq 20$, $0 \leq x_3 \leq 15$, $0 \leq x_4 \leq 25$, $0 \leq x_5 \leq 20$, $0 \leq x_6 \leq 20$.

Remark (why equalities at the plants). Physically a plant may ship *up to* its stock, so $x_1 + x_3 \leq 30$ is also a correct constraint. Here the two versions have the same feasible set: adding the shop and depot equations gives $(x_1 + x_3) + (x_2 + x_6) = 50$, and with $x_1 + x_3 \leq 30$, $x_2 + x_6 \leq 20$ both must be tight. Whenever total supply equals total demand, $\leq$ and $=$ at the supply nodes coincide. If supply exceeded demand, use $\leq$ at the plants, or keep equalities and add a dummy demand node absorbing the surplus at zero cost.

- (c) Routes to S_1 : P_1DS_1 at 4, P_2DS_1 at 5, P_1S_1 at 6. Routes to S_2 : P_1DS_2 at 5, P_2S_2 at 5, P_2DS_2 at 6. Plan: $x_1 = 30$, $x_4 = 25$, $x_5 = 5$, $x_6 = 20$, $x_2 = x_3 = 0$. Balances: P_1 : 30; P_2 : 20; D : $25 + 5 = 30$; S_1 : 25; S_2 : $5 + 20 = 25$. Capacities: $30 \leq 30$, $25 \leq 25$, $5 \leq 20$, $20 \leq 20$. Cost = $60 + 50 + 15 + 100 = 225$ (hundreds of MAD).
- (d) Dropping supplies and capacities, each tonne for S_1 still travels some route to S_1 , costing at least 4; each tonne for S_2 costs at least 5. So every feasible flow costs at least $25 \cdot 4 + 25 \cdot 5 = 225$. The plan achieves 225: optimal.
- (e) Total supply would be $45 < 50 =$ total demand. Summing the balance equations then gives $0 = 45 - 50 \neq 0$, impossible; no feasible flow exists.

Solution 5

- (a) Minimise $\sum_{i,j} c_{ij}x_{ij}$ subject to $x_{A1} + x_{A2} + x_{A3} = 1$ (and likewise for B, C), $x_{A1} + x_{B1} + x_{C1} = 1$ (and likewise for incidents 2, 3), $x_{ij} \in \{0, 1\}$.
- (b) Cheapest pair: $(A, 1)$ at 1. Remaining cheapest: $(C, 3)$ at 3. Then $(B, 2)$ at 4. Greedy total 8.
- (c) $A1B2C3$: 8; $A2B1C3$: 7; $A1B3C2$: 19; $A2B3C1$: 20; $A3B1C2$: 20; $A3B2C1$: 22. Optimum 7 ($A \rightarrow 2$, $B \rightarrow 1$, $C \rightarrow 3$). Greedy is not optimal.
- (d) Dropping the three “each incident” constraints lets every analyst pick their own cheapest incident independently; the relaxed optimum is the sum of row minima $1 + 2 + 3 = 6$, a

lower bound. It does not match 7, so it does not prove optimality.

- (e) For any assignment x , $\sum c_{ij}x_{ij} \geq \sum (u_i + v_j)x_{ij} = \sum_i u_i \sum_j x_{ij} + \sum_j v_j \sum_i x_{ij} = \sum_i u_i + \sum_j v_j$, using $x \geq 0$ and the two families of constraints. Take $u = (1, 2, 3)$, $v = (0, 1, 0)$: check $c_{A2} = 2 \geq 2$, $c_{B2} = 4 \geq 3$, $c_{C2} = 9 \geq 4$, and all other entries against $u_i + 0$. The bound is $6 + 1 = 7$, matching (c): optimal.

Solution 6

- (a) Count pairs (node, arc) with the node the tail of the arc: each node i contributes $|\delta^+(i)|$, each arc contributes exactly one tail. Same with heads.
- (b) Sum $x(\delta^+(i)) - x(\delta^-(i))$ over all nodes: the total is 0 (each arc cancels). With conservation at every node except t , the sum reduces to $x(\delta^+(t)) - x(\delta^-(t)) = 0$. Since the value equals $x(\delta^-(t)) - x(\delta^+(t))$ (Lemma 1.6.11), the value is 0.
- (c) $\text{Value}(x) = x(\delta^+(S)) - x(\delta^-(S)) = u(\delta^+(S)) - 0$. For any flow x' , $\text{value}(x') \leq u(\delta^+(S))$ by the cut bound. So x attains an upper bound valid for all flows: it is maximal.
- (d) Sum all constraints: the left sides sum to 0. Supply rows give $\leq b_i$, demand rows give $= b_i$ (negative). Hence $0 \leq \sum_{\text{supply}} b_i + \sum_{\text{demand}} b_i$, i.e. total supply $\geq$ total demand.